

Monitoring the progression and pre-clinical diagnosis of Parkinson's disease by detection of α -synuclein oligomers in CSF using the Seed Amplification Assay (SAA).

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PPMI Project ID: [155]

Summary

The α -synuclein (α Syn) Seed Amplification Assay (α S-SAA) takes advantage of the intrinsic self-replicating nature of α Syn misfolded aggregates (seeds) and multiplies them for their detection in synucleinopathy cases such as Parkinson's disease (PD), multiple system atrophy (MSA), and dementia with Lewy bodies (DLB). Amplification occurs *in vitro* when α Syn seeds in cerebrospinal fluid (CSF) are fragmented to generate numerous smaller seeds that grow by incorporating native recombinant protein. Given the nature of the technique, minute quantities of α Syn seeds in human bodily fluids can be multiplied and thereafter detected by conventional methods, such as immunoassays or fluorescence. Following this principle, α S-SAA was standardized for the highly sensitive and specific detection of α Syn seeds in CSF from patients with PD and other synucleinopathies¹⁻³.

Method

Amprion- α S-SAA (formerly named "Amprion fast α S-PMCA" in LONI).

The reaction mixture for amplification of endogenous α Syn seeds in CSF has a final volume of 200 μ L and consists of 100mM PIPES pH 6.5, 0.5M NaCl, 10 μ M ThT, 0.3mg/mL recombinant α Syn, and 40 μ L CSF. 1x 3/32" silicon nitride bead blocked with 1% BSA¹ is included per reaction. The reaction is performed in 96-well plates and the samples are analyzed in 3 horizontal replicates (kinetic parameters are reported per replicate (Rep1, Rep2, Rep3)). The plate is incubated at 37°C and shaken for 1 min every 29 min. Fluorescence is read at 440nm-Ex/490nm-Em as relative fluorescence units (RFU) after every shaking cycle for 300 cycles (150h, 6.2 days). Under these conditions, CSF samples containing α Syn seeds display robust aggregation curves in all replicates within 150h, while CSF samples from control patients do not. This runtime is significantly shorter than the original assay described by Shahnawaz and collaborators⁴, which took 300h and was previously uploaded as "Amprion slow α S-PMCA" to the LONI database. In this project, **1,410** unidentified CSF samples were analyzed over 4 months using a single lot of recombinant α Syn substrate and 3 FLUOstar Omega shaker/readers (IDs 2, 4, and 6). Samples were randomized by the PPMI bio repository to control for each group of samples to be analyzed over different runs and different instruments. Each sample was also assigned to certain wells in a plate.

Data analysis

Amprion- α S-SAA Qualitative.

α Syn amplification normally presents a classic amyloid aggregation curve in the “Amprion- α S-SAA”, including a lag phase, an exponential growth, and a final plateau. Samples without aggregation display background fluorescence values, but no aggregation curve. Amprion developed an algorithm to make a final determination of samples using the information from all 3 replicates, which has been reported in detail elsewhere in main text² and supplemental materials¹. Samples in which α Syn seeds are detected are called “Positive”, while samples where seeds are not detected are called “Negative” (QUALAlgo). In borderline cases where a definitive call cannot be made based on the three replicates, the sample is deemed “Inconclusive”, and the sample is recommended to be reanalyzed. Both inconclusive and the conclusive retest results are reported. “Retest” in the Comments column indicates the retest results for a given sample. Since data is requested to be reported per replicate, qualitative results are included for all 3 replicates (QUALAlgoRep1, QUALAlgoRep2, QUALAlgoRep3) (Table I). Thus, qualitative results for Rep1-3 are the same since the algorithm uses the information from all 3 replicates to make a final determination.

α S-SAA amplification parameters.

Fluorescence traces were further analyzed to estimate kinetic parameters that characterize the amplification curve of each replicate, such as the maximum fluorescence (Fmax [RFU]), the time to reach 50% of Fmax (T50 [hours]), the time to threshold (TTT [hours]), the slope [RFU/s], and the area under the curve (AUC, [%*s]) (Table I).

Reported exceptions.

Some vials did not have enough volume and the last replicate was analyzed with less than 40 μ L of sample, which is reported as an exception for each replicate (SampleVolRep1, SampleVolRep2, SampleVolRep3). Values for SampleVol: “NEV1” (Not Enough Volume 1) for CSF samples that contained between 20-30 μ L of the volume required for a replicate; “NEV2” (Not Enough Volume 2) for CSF samples that contained more than 30 μ L but less than 40 μ L; and “NEV3” (Not Enough Volume 3) is for CSF samples that contained 10 μ L or less.

Table I. Description of TESTNAMES and outcome for the data uploaded to LONI.

Column	Outcome
Amprion- α S-SAA Qualitative (QUALAlgoRepX ¹)	Character: Positive, Negative, Inconclusive. All replicates have the same value.
Amprion- α S-SAA Fmax (FmaxRepX ¹)	Numeric [RFU]. Each replicate has a different value.
Amprion- α S-SAA T50 (T50RepX ¹)	Numeric [hours]. Each replicate has a different value. Replicates without amplification do not have a T50 value (-).
Amprion- α S-SAA TTT (TTTRepX ¹)	Numeric [hours]. Each replicate has a different value. Replicates without amplification do not have a TTT value (-).
Sample volume (SampleVolRepX ¹)	Character: “NEV1” (20 to 30 μ L), NEV2 (30 μ L to <40 μ L), NEV3 ($\leq$ 10 μ L). Indicates the replicate did not have the 40 μ L usually used per reaction.
Amprion- α S-SAA AUC (AUCRepX ¹)	Numeric [%*s]. Each replicate has a different value. Replicates without amplification do not have an AUC value (-).
Amprion- α S-SAA Slope (SLOPERepX ¹)	Numeric [RFU/s]. Each replicate has a different value. Replicates without amplification do not have a Slope value (-).
Amprion- α S-SAA Instrument (InstrumentRepX ¹)	Numeric. Instruments were identified as 2, 4 and 6.

¹RepX represents Rep1, Rep2, and Rep3.

Results

Samples were received and analyzed deidentified. Of the 1,410 CSF samples analyzed, there were 1,371 samples (97.2%) with a conclusive result and 39 samples (2.8%) with inconclusive result. There were 2 samples that had completely empty tubes provided; therefore, these CSF samples were not analyzed, and no α S-SAA results are reported. Of the Inconclusives, 17 samples did not have sufficient volume for retest (Inconclusives/not-retested), but the other 22 Inconclusive samples were retested. Of the 22 retests, 15 produced a conclusive result (71.4%), while 6 produced inconclusive results once again. Including retest results as final, there were 761 (54%) Positives, 625 (44.3%) Negatives, 17 (1.2%) Inconclusives/not-retested, and 7 (0.5%) Inconclusives/retested.

References

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4. Shahnawaz, M. *et al.* Development of a Biochemical Diagnosis of Parkinson Disease by Detection of α -Synuclein Misfolded Aggregates in Cerebrospinal Fluid. *JAMA Neurol.* **74**, 163 (2017).

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